

GlobalCommerce

Enterprise cross-border digital commerce platform

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EduQual Level 6 Diploma in Artificial Intelligence Operations

Local working prototype with simulated external commerce services

Business problem

Separate records make international commerce harder to control.

Stock uncertainty	Businesses can accept orders for units already promised.
Cross-border complexity	Currencies and landed costs complicate checkout.
Limited delivery visibility	Teams struggle to explain where an order stands.
Uncertain demand	Excess stock ties up money. Shortages lose sales.

Project scope and presentation roadmap

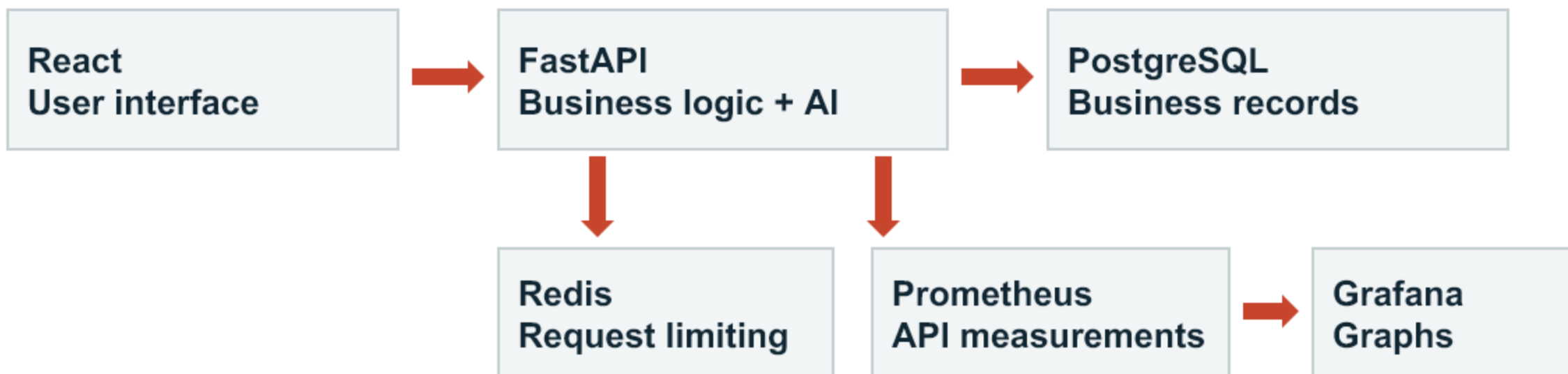
Five business modules share one local application and database.

Module	Evidence in this presentation
Marketplace management	Seller verification and customer access
Inventory management	Warehouse balances and stock movements
International orders and logistics	Reservation, charges, delivery and cancellation
AI-assisted demand forecasting	History, baseline, error and replenishment
Executive analytics	Financial definitions and operational exports

Roadmap: business problem, architecture, evidence, limitations, live demonstration

System architecture

A modular backend handles business rules and forecasting.



Docker Compose runs six local services. Responses return to the interface.

Marketplace and access control

The server checks the user's role and ownership of records.

Actor	Example responsibility or boundary
Administrator	Verify sellers and manage accounts and catalogue
Merchant	Manage owned products and view own sales report
Customer	Create orders and access own order records
Warehouse and logistics staff	Perform permitted stock and shipment operations

A verified seller can supply active products. A hidden button alone is not security.

Inventory rules

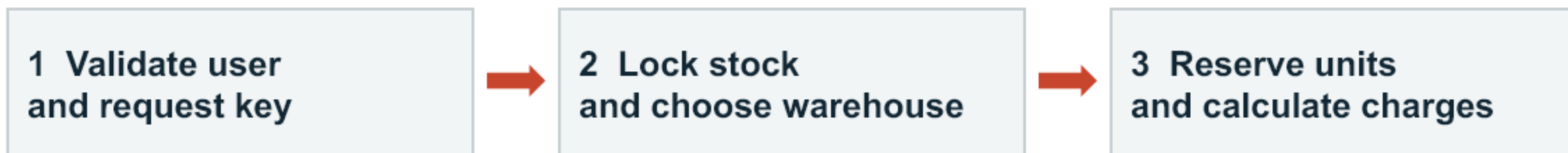
Available stock = on hand – reserved

Illustrative stage	On hand	Reserved	Available
Before order	100	0	100
After reserving 2 units	100	2	98
After delivering those units	98	0	98

Illustrative figures. This prototype deducts on-hand stock at delivery.

Order transaction and safe retries

The backend commits the order and reservation together.



Commit

Order, payment simulation, shipment and audit record

Failure / retry

Roll back an error. Return the same order for a matching retry.

One warehouse must supply the whole order. Allocation favours available quantity.

International checkout

The order records its currency, exchange rate and charge components.

Component	Illustrative GBP amount
Goods	100.00
Shipping	12.00
Duty: 5% of goods	5.00
Tax: 8% of goods plus duty	8.40
Total checkout value	125.40

Illustrative component arithmetic. Rates and charges are simulated, not live trade rules.

Logistics visibility

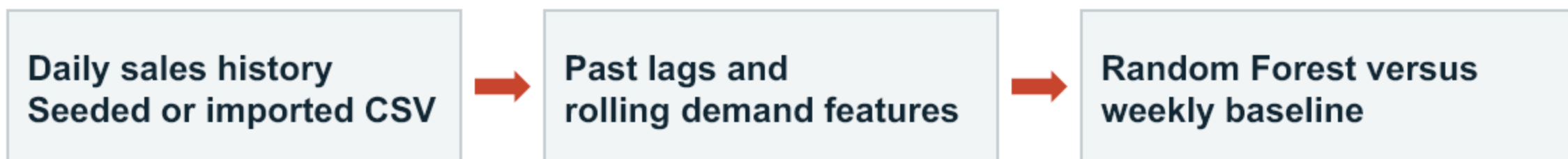
A tracking reference and event history connect each shipment to its order.

Normal progression	What the operator can explain
Created, then picked	The warehouse prepares the order
In transit, then customs	The shipment progresses across borders
Out for delivery, then delivered	The final delivery completes the workflow
Delay and recovery	Valid transitions preserve the event history

Carrier events and payment capture are simulated. The API rejects repeated delivery.

Demand forecasting method

Past sales inform a future estimate for each product and warehouse.



Earlier 80% trains

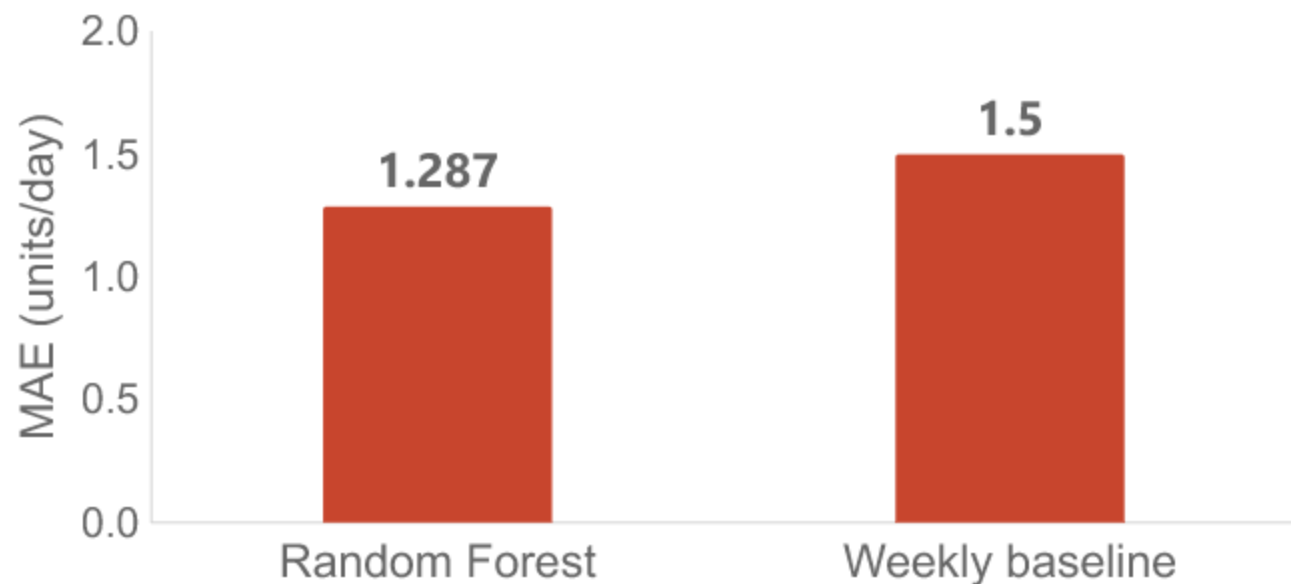
Later 20% evaluates

The lower holdout MAE selects the forecast model.

Orders do not automatically update training history. The same holdout also selects the model.

Recorded forecast evidence

NT-NOISE-01 / DXB-01: one product and warehouse in the synthetic test dataset



Lower error is better

96 training days
24 holdout days

Forest RMSE: 1.453
Selected: Random Forest

Saved run: 6 September 2026. Synthetic data does not establish commercial accuracy.

Replenishment and management decisions

A person reviews each stock recommendation before acting.

$$60 + 15 - 50 = 25 \text{ units}$$

Forecast demand + safety stock – available stock

Illustrative recommendation: receive 25 additional units

Floor at zero. Stock at forecast time may differ from current stock.

Analytics and monitoring

Business reports and service measurements answer different questions.

Business analytics	Technical monitoring
Orders and units	API requests per second
Goods value versus checkout total	p95 response time
Available stock by warehouse	Response codes and server errors
Delivery completion and CSV exports	Prometheus measurements and Grafana graphs

Delivery completion differs from on-time delivery. API requests differ from orders.

Security and failure handling

Controls protect the workflow, with explicit prototype limitations.

Unauthorized access	JWT, password hashes and server-side ownership checks
Conflicting stock updates	Transactions, locks and validated lifecycle transitions
Interrupted requests	Stable retry keys and checks of persisted order state
Service or data failure	Health checks, logs, backups and a tested restore copy

Production still needs TLS, MFA and managed secrets. Redis failure permits requests.

Verification evidence and boundaries

Recorded results support specific claims, rather than complete correctness.

Evidence	Result and scope
Backend regression	20 passed, 6 September 2026
Application line coverage	89.91%: 1,337 of 1,487 lines
PostgreSQL concurrency	Duplicate requests, competing stock and delivery
Forecast evaluation	9 synthetic series, dated model and baseline errors

No production load test. Full container OS scan and final live rebuild remain unverified.

AI assistance and prototype limits

AI supports the work. The presenter must explain and defend every claim.

AI assistance

Code and configuration drafting
Documentation and slide preparation
Debugging and test assistance

Current boundaries

Simulated payments and carrier events
Synthetic or imported sales history
Local deployment with demo settings
No measured enterprise-scale capacity

Personal ownership means understanding decisions and reproducing the checks.

Live demonstration

One customer orders two units. Every step has a visible result.

Time	Action	Visible evidence
0:00–0:40	Starting stock	On hand, reserved and available
0:40–2:40	Order and delivery	Order ID, charges, stock changes and history
2:40–3:30	Forecasting	Model error and stock recommendation
3:30–4:15	Pre-dispatch cancellation	A separate order releases reserved units
4:15–5:00	Analytics and monitoring	Business totals versus API measurements

Use the EXE launcher before the exam. Keep the slides and website open.

Conclusion

The prototype connects the core commerce workflow.

Demonstrated value	Linked orders and inventory reduce conflicting records.
Decision support	Forecast comparisons inform replenishment decisions.
Operational evidence	History, reports and monitoring make behaviour inspectable.
Next stage	Real integrations, stronger security and measured scaling.

Appendix: architecture trade-offs

Decisions fit a local prototype. Production changes need evidence.

Decision today	Reason	Future extension
One modular API	Simple deployment and shared transactions	Forecast worker when load justifies it
One warehouse per order	Simpler allocation and shipment consistency	Split fulfilment and multi-shipment state
Deduct stock at delivery	Clear assessment lifecycle	Warehouse and in-transit stock ledgers
Human-reviewed forecasts	Avoid unchecked automated purchasing	Lead times, approval and calibrated policies

Appendix: evidence and assessment references

Project evidence

reports/verification/README.md
docs/model-evaluation.md
docs/requirements-traceability.md
diagrams/index.html
AI_USAGE.md

Assessment references

Supplied oral presentation exam guide
Assigned project topic document
Sample oral exam: Talha Nawaz
YouTube: tMiNZTVoy5c

Presentation: 25 marks. Interview: 25 marks. Slides alone cannot guarantee a score.